

Crystal structure of the MOP flippase MurJ in an inward-facing conformation

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Peptidoglycan (PG) protects bacteria from osmotic lysis, and its biogenesis is a key antibiotic target. A central step in PG biosynthesis is flipping of the lipid-linked PG precursor lipid II across the cytoplasmic membrane for subsequent incorporation into PG. MurJ, part of the multidrug/oligosaccharidyl-lipid/polysaccharide (MOP) transporter superfamily, was recently shown to carry out this process. However, understanding of how MurJ flips lipid II, and of how MOP transporters operate in general, remains limited due to a lack of structural information. Here we present a crystal structure of MurJ from *Thermosiphon africanus* in an inward-facing conformation at 2.0-Å resolution. A hydrophobic groove is formed by two C-terminal transmembrane helices, which leads into a large central cavity that is mostly cationic. Our studies not only provide the first structural glimpse of MurJ but also suggest that alternating access is important for MurJ function, which may be applicable to other MOP superfamily transporters.

The building block of PG, GlcNAc-MurNAc-pentapeptide, is synthesized in the cytosol and attached to the inner leaflet of the cytoplasmic membrane as the lipid-linked precursor undecaprenyl diphosphate-MurNAc-pentapeptide-GlcNAc, known as lipid II (refs. 1,2). Subsequently, lipid II is flipped from the cytoplasmic side to the periplasmic side of the membrane for polymerization into PG^{3,4}. Although the precursor synthesis and polymerization steps of PG biogenesis have been well studied, lipid II flipping is poorly understood, and flippase candidates MurJ, Amj, and FtsW have been identified only recently^{5–9}.

Recent studies strongly suggest that MurJ is the long-sought-after lipid II flippase^{7,9,10}. Depletion of MurJ (formerly known as MviN) results in the accumulation of PG precursors, a decrease in PG biosynthesis, and ultimately cell lysis in *Escherichia coli* and *Burkholderia cenocepacia*^{7,9,11}. Notably, MurJ-dependent flipping of radiolabeled lipid II was detected in *E. coli* cells and spheroplasts¹⁰. Furthermore, the FtsW paralog RodA was recently characterized as a PG glycosyltransferase¹².

MurJ belongs to the MVF (mouse virulence factor) family of MOP superfamily transporters, which also includes the MATE (multidrug and toxic compound extrusion) transporter and eukaryotic OLF (oligosaccharidyl-lipid flippase) families¹³. In addition to the canonical MOP transporter core consisting of 12 transmembrane helices (TMs), MurJ has two additional C-terminal TMs (13 and 14) of unknown function. The outward-facing conformation of *E. coli* MurJ has been probed extensively by homology modeling, the *in vivo* substituted-cysteine accessibility method, and mutagenesis studies, which revealed a solvent-exposed cavity that is essential for function^{14,15}. However, a complete understanding of how MurJ can flip such a large (~1,900 Da) and negatively charged lipid substrate remains elusive because of the lack of crystal structures. Another intriguing question

is whether MurJ-dependent lipid II flipping occurs via an alternating-access mechanism; recent studies have suggested that alternating access is not used for lipid-linked oligosaccharide flipping by the ABC transporter PglK¹⁶. Furthermore, among the MOP transporter superfamily, only MATE-family transporters have been crystallized so far, all in outward-facing conformations^{17–21}. As a result, the molecular mechanism underlying alternating-access transport by MOP transporters is still poorly understood.

To understand the structural basis of lipid II flipping by MurJ, we carried out structural and mechanistic studies of MurJ. We report the crystal structure of MurJ. Our structural analyses, combined with mutagenesis studies, have allowed us to propose a model of lipid II flipping by MurJ and provide a foundation for understanding MurJ function at an atomic level.

RESULTS

Overall architecture of MurJ

We solved the structure of MurJ from the hyperthermophilic bacterium *T. africanus* (MurJ_{TA}) to 2.0-Å resolution from crystals grown in lipidic cubic phase. We determined experimental phases to 3.5-Å resolution by single-wavelength anomalous dispersion from selenomethionine-substituted crystals (Supplementary Fig. 1). The electron density map allowed residues 4–470 (out of 475 residues) to be built unambiguously, and the final model was refined to good geometry (Table 1). We deduced the locations and identities of ions in the structure from results of anomalous scattering experiments (Supplementary Fig. 2 and Supplementary Table 1).

Our structure of MurJ_{TA} is strikingly different from existing MATE structures. In the MurJ_{TA} structure, the N lobe (TMs 1–6) and C lobe (TMs 7–14) are arranged in an inward-facing N-shape conformation (Fig. 1a,b), rather than the outward-facing V-shape

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Table 1 Data collection and refinement statistics

	Native (PDB 5T77) ^a	SeMet ^b
Data collection		
Space group	C2	C2
Cell dimensions		
a, b, c (Å)	94.16, 99.57, 74.66	93.86, 99.74, 75.67
α, β, γ (°)	90, 112.91, 90	90, 101.84, 90
		Peak
Wavelength (Å)	0.9791	0.9791
Resolution (Å)	100–2.00 (2.07–2.00) ^c	70–3.15 (3.26–3.15) ^c
R _{pim} (%)	6.5 (55.0)	4.2 (25.0)
I/σ(I)	9.47 (1.06)	30.8 (4.10)
CC _{1/2}	0.99 (0.50)	1.00 (0.87)
Completeness (%)	98.0 (94.0)	99.5 (95.0)
Redundancy	4.2 (2.2)	50.4 (30.5)
Refinement		
Resolution (Å)	34.2–2.00 (2.07–2.00)	
No. reflections	41,992 (4,018)	
R _{work} /R _{free} (%)	18.9/21.4	
No. atoms	4,404	
Protein	3,743	
Mono-olein/PEG/ Cl [−] /Zn ²⁺ /Ca ²⁺ /Na ⁺	453/48/5/1/3/2	
Water	149	
B factors		
Protein	38.2	
Mono-olein/PEG/ Cl [−] /Zn ²⁺ /Ca ²⁺ /Na ⁺	68.5	
Water	43.2	
R.m.s. deviation		
Bond lengths (Å)	0.003	
Bond angles (°)	0.55	

^aMerged from three native crystals grown in lipidic cubic phase. ^bMerged from 12 selenomethionine (SeMet) crystals grown in lipidic cubic phase. ^cValues in parentheses are for the highest-resolution shell. PEG, polyethylene glycol.

conformation observed in all existing MATE transporter structures^{17–21}. The cytoplasmic orientation of the termini and the positive-inside-surface electrostatics (**Supplementary Fig. 1e**) support the inward-facing topology. Although strong two-fold pseudo-rotational symmetry is observed between the two lobes in the MATE structures^{17,19,20}, the symmetry between the N lobe and the core C lobe (TMs 7–12) of MurJ_{TA} is notably distorted (**Fig. 1c**). In particular, the conformations of TM 1 and TM 2 are very different from those of the symmetry-related TM 7 and TM 8 (**Fig. 1c**). TM 1 extends out from the transporter core instead of interacting with the C lobe as in other MATE structures, and TM 2 is broken at the Gly(Ala)56-Glu57-Gly58-Ala59 motif that is conserved in Gram-negative bacteria (**Supplementary Note 1**), allowing this region to protrude into the central cavity. In addition, both lobes of the core transport domain of MurJ_{TA} are substantially wider and shorter than those in MATE transporters (**Fig. 1c**).

The hydrophobic groove and central cavity of MurJ_{TA}

C-terminal TMs 13 and 14 extend out from the transporter core and are oriented beside TM 9. These two additional TMs, absent in MATE proteins, form a curved groove (**Fig. 2a**) that is ~20 Å long. This groove is lined mostly with hydrophobic residues from TMs 1, 8, 9, and 14. The hydrophobic groove penetrates into the central transport cavity via a membrane portal located between TMs 1 and 8 (**Fig. 2b**).

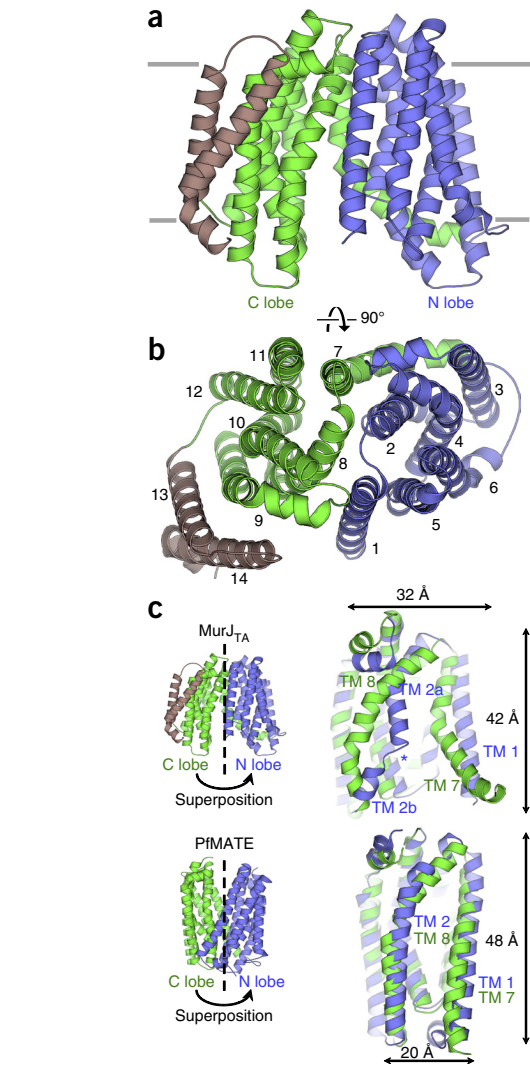


Figure 1 Architecture of MurJ_{TA}. **(a)** View from within the membrane. The N lobe (TMs 1–6) and C lobe (TMs 7–12) of the core transport domain are shown in blue and green, respectively; the additional TMs 13 and 14 are shown in brown. **(b)** Extracellular view of MurJ_{TA} showing the arrangement of the N lobe (TMs 1–6; blue) and C lobe (TMs 7–12; green). **(c)** Structural superposition between the N lobe (blue) and C lobe (green) of MurJ_{TA} and of PfMATE (PDB 3VVN), showing the broken symmetry in MurJ_{TA}, in particular between TMs 1 and 7 as well as between TMs 2 and 8. The core transport domain of MurJ_{TA} is also shorter and wider than that of PfMATE. The helix-breaking Gly56-Glu57-Gly58-Ala59 motif in TM 2 is denoted by the blue asterisk.

Most of the residues lining the portal have small side chains, with the notable exception of Arg18, which is essential for *E. coli* MurJ function¹⁵. Beyond the portal, the groove opens up into a large inward-facing central cavity ~20 Å wide and penetrating up to ~20 Å into the membrane (**Fig. 2c**). This cavity can be subdivided into a proximal site (directly connected to the hydrophobic groove) and a distal site (far from the groove). The proximal site is highly positively charged (**Fig. 2d**), mainly owing to the presence of Arg24, Arg52, and Arg255, counterparts of which were found to be important for the function of *E. coli* MurJ¹⁵. In our structure, a chloride ion, identified by anomalous signal from a bromide-soaked crystal, is located inside the proximal site (**Fig. 2c** and **Supplementary Fig. 2b**). Whether this chloride ion is physiologically relevant remains unknown; the chloride might be nonspecifically bound

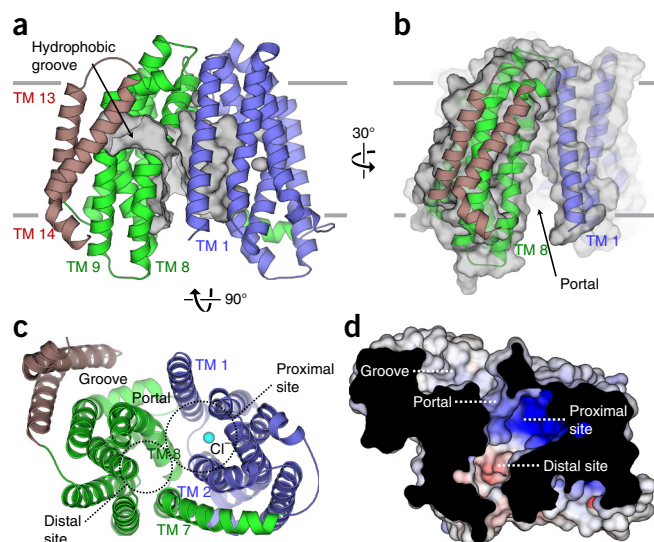


Figure 2 The hydrophobic groove of MurJ_{TA} is linked to the central cavity. (a) TMs 13 and 14 (brown) form a hydrophobic groove (surface shown in gray). (b) The hydrophobic groove leads into the central cavity through a portal between TMs 1 and 8. (c) The central cavity is formed mainly by TMs 1, 2, 7, and 8. This large cavity can be subdivided into proximal and distal sites with respect to the groove. A chloride ion is found in the proximal site (cyan sphere). (d) The proximal site is strongly cationic, whereas the distal site is mildly anionic.

to MurJ as a result of the high concentration of chloride in the crystallization drop (containing 400 mM CaCl₂).

Model of lipid II binding to MurJ_{TA}

Although we observed unassigned electron density peaks at both the membrane portal (Supplementary Fig. 3a) and the distal site of the central cavity (Supplementary Fig. 3b), the quality of the density peaks was not sufficient for their unambiguous assignment. We therefore used a ligand-docking approach to test whether the binding of lipid II to our MurJ_{TA} structure was feasible. In the docking result with the highest score, lipid II was docked into the groove and cavity with good geometry and feasible electrostatic interactions (Fig. 3a). In this model, the undecaprenyl tail of lipid II associates with the hydrophobic groove, the diphosphate and sugar moieties are placed in the proximal site, and the pentapeptide moiety is located in the distal site of the cavity. In particular, the unassigned electron density at the distal site was where the pentapeptide moiety was located in our docking result. On the basis of this observation, we were able to build the pentapeptide moiety partially into the electron density (Supplementary Fig. 3c), but it was excluded from refinement because of the uncertainty of our assignment.

Guided by our docking result, we next tested conserved residues in the portal and central cavity for their importance to lipid II binding by carrying out functional complementation assays as previously described^{7,11} (Fig. 3b). We used *E. coli* strain NR1154, in which the gene expressing endogenous MurJ (*murJ_{EC}*) is under the control of arabinose⁷. The introduction of a plasmid containing *murJ_{TA}* under the control of an isopropyl β-D-thiogalactopyranoside (IPTG)-inducible promoter allowed growth of this strain in the absence of arabinose, provided that IPTG was present (Fig. 3b). Mutation of conserved residues (Ser17, Arg18, Arg24, Arg52, and Arg255) at the proximal site failed to complement MurJ_{EC} function, consistent with the idea that these residues are important for recognizing the diphosphate and/or sugar moieties of lipid II (Fig. 3a).

Occlusion of the membrane portal (via mutation of Leu259 to tryptophan) or changes in the shape of the groove (caused by mutation of Phe256 to alanine) also led to loss of complementation, suggesting disruption of lipid II binding (Fig. 3c). As failure to complement could alternatively be due to loss of protein expression, we analyzed total membrane fractions by western blotting. All mutant MurJ_{TA} proteins were expressed except D235A and N374A, which we removed from our analysis (Supplementary Fig. 4).

We next analyzed *in vivo* chemical genetics data from previous studies in the context of our structure to gain further insight into lipid II binding. Ruiz and co-workers¹⁰ found that although a single-cysteine mutant of *E. coli* MurJ (MurJ_{EC} A29C) was functional, addition of the membrane-impermeant cysteine-reactive agent 2-sulfonatoethyl methanethiosulfonate (MTSES) inhibited flippase activity and led to cell lysis. They identified a total of five positions that showed a similar MTSES-dependent phenotype in MurJ_{EC}^{10,14}. Because we found that MurJ_{TA} can complement MurJ_{EC} and that the essential residues are conserved between the two, we mapped these residues to the structure of MurJ_{TA}. Three of the residues (Phe49, Ser254, and Leu258) are located at the junction between the proximal and distal sites in the central cavity (Fig. 3d) but are not strongly conserved (Supplementary Note 1), which suggests that they do not directly participate in lipid II binding. Consistent with our model, mutation to cysteine at these positions would not affect lipid II binding, but the addition of a bulky adduct from MTSES would be likely to obstruct lipid II binding (Fig. 3d). However, we caution against excessive interpretation of our docking and complementation results, as these experiments are not direct measurements of lipid II binding. We performed these experiments to refine our hypothesis regarding the likely locations of the different chemical moieties of lipid II, not to propose a definitive structural model of lipid II bound to MurJ.

Importance of alternating access for lipid II flipping by MurJ

Because our MurJ_{TA} structure is the first structure of a MOP transporter captured in an inward-facing conformation, it offers an opportunity to study the possibility of an alternating-access transport mechanism. We observed from our inward-facing structure that TM 7 is bent by ~45° about Ser228, in contrast to its straightness in the outward-facing MATE structures. As most MurJ and MATE sequences have either a serine or a threonine residue in this region (Supplementary Note 1), we hypothesized that bending of TM 7 about this residue might be relevant for the transition to the outward-facing state. We generated an outward-facing model of MurJ_{TA} (Supplementary Data Set 1) by splitting the model at the center of TM 7 (Ser228) and aligning the N-terminal fragment (TMs 1–7a) and the C-terminal fragment (TMs 7b–12) as rigid bodies with the structure of PfMATE (PDB 3VVN), which we chose because it is the highest-resolution MATE structure published²⁰. The resulting model aligned better at the C lobe than at the N lobe. Because of the substantial structural difference between MurJ and MATE transporters, our outward-facing model generated on the basis of the MATE structure should be interpreted with caution.

To test the validity of our outward-facing model, we mapped those five positions that are functionally sensitive to MTSES¹⁰ to the inward-facing structure and outward-facing model. All five positions are accessible from the periplasm in the outward-facing model but not in the inward-facing structure, showing that our model is consistent with the MTSES data (Fig. 4a,b). Notably, mapping of two MTSES-sensitive positions located at the periplasmic end of TM 1 (Ala29) or TM 8 (Ser248) suggests that alternating access is important for lipid II

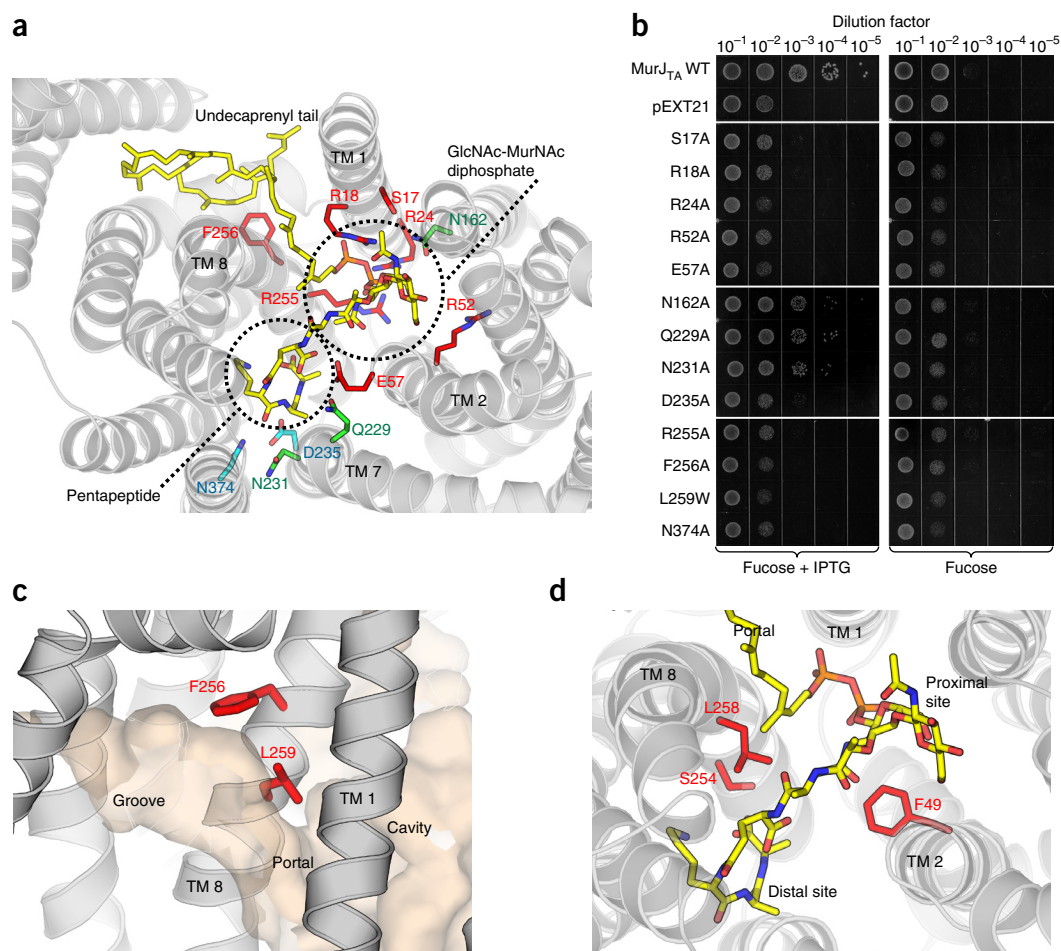


Figure 3 Model of lipid II binding to MurJ_{TA}. **(a)** A model of lipid II (yellow sticks) docked into the structure of MurJ_{TA}. Each chemical moiety of lipid II is indicated. Residues that were mutated to alanine and assayed for complementation in **(b)** are shown as sticks. Residues essential for the function of MurJ_{TA} are colored red, nonessential residues are colored green, and residues whose replacement with Ala resulted in lack of expression are shown in cyan. **(b)** Functional complementation data for MurJ_{TA} in *E. coli* NR1154. Cells were transformed with plasmids encoding MurJ_{TA} (wild-type or mutant) or without insert (pEXT21) and were subsequently depleted of endogenous MurJ by serial dilution on plates containing the anti-inducer fucose in the absence of arabinose; MurJ_{TA} expression was induced by addition of IPTG. Data shown are representative of three technical replicates. **(c)** Occluding the portal (with L259W) or perturbing the groove surface (with F256A) results in a protein that is unable to complement MurJ activity. **(d)** Mapping of MTSES-sensitive positions from *E. coli* MurJ (shown as red sticks), for which cell lysis or cell shape defects were observed upon MTSES treatment of cysteine mutants¹⁰, to the structure of MurJ_{TA}. These residues (F49, S254, and L258) do not seem to interact with the diphosphate or sugar groups and are not strongly conserved (Supplementary Note 1). Other residues represented by sticks are color-coded as in **(a)**.

flipping. In our inward-facing MurJ structure, these positions are located at the interface between the N and C lobes, and thus have important roles in occluding access to the periplasmic side. It is likely that MTSES adduct formation at these sites disrupted the inward-facing state of MurJ, thereby trapping MurJ in the outward-facing state, resulting in disruption of lipid II flipping and cell lysis (Fig. 4c,d). An alternative hypothesis whereby the adducts occlude the outward-facing cavity could be ruled out because mutations of these sites to large polar or charged amino acids (but smaller than the adduct) other than cysteine also led to cell lysis¹⁰. These data suggest that both the inward-facing and the outward-facing state exist for MurJ, and that alternating access might be important for lipid II flipping.

DISCUSSION

Although MATE proteins have been characterized as catalyzing either Na⁺- or H⁺-coupled antiport transport of substrates^{17–20},

it is unclear whether MurJ requires any counter-ion for lipid II flipping. Unfortunately, our structure does not provide an answer to this question. Most of the ions observed in the structure are unlikely to function as the counter-transporting ions for lipid II flipping, as they are located outside of the central cavity and are coordinated mostly by water molecules (Supplementary Fig. 2). This does not rule out the involvement of Na⁺ or H⁺, as they are not only difficult to identify but also likely to be bound only in the outward-facing state. However, the chloride ion in the central cavity and a zinc ion at the beginning of TM 7 might be functionally important. The chloride ion (Cl[−] 1; Supplementary Fig. 2b) is located in the proximal site near Arg24 and Arg255, which were critical for the function of both MurJ_{TA} and *E. coli* MurJ^{14,15}. The putative zinc ion is coordinated by two histidine residues and two chloride ions (Supplementary Fig. 2c–e), which is surprising, because at no point was zinc added during the protein purification or crystallization process. Because

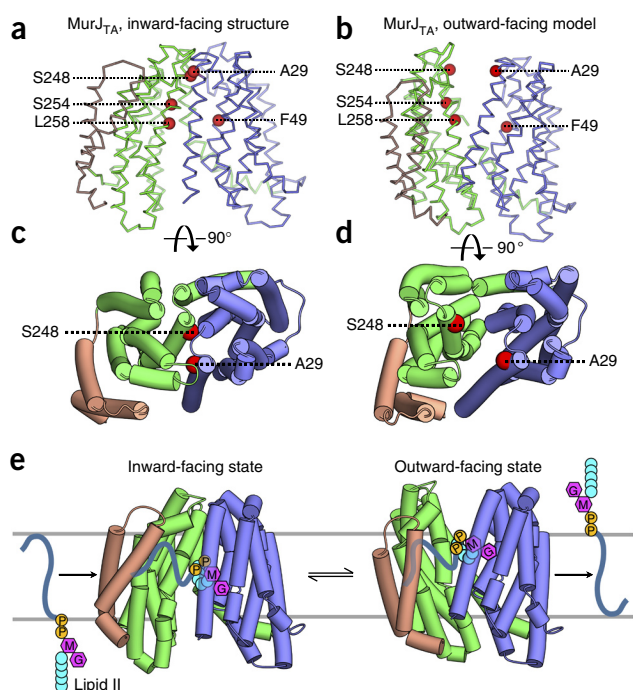


Figure 4 Putative mechanism of lipid II flipping by MurJ. Previously reported MTSES-sensitive positions (red spheres) that resulted in loss of MurJ function after MTSES treatment¹⁰ were mapped to the structure of MurJ_{TA}. (a) These residues are not accessible to the membrane-impermeable agent MTSES in the inward-facing configuration. (b) In the outward-facing model (Supplementary Data Set 1), all of these residues are exposed to the periplasm, explaining the MTSES sensitivity of cysteine substitutions at these positions. (c,d) Both Ala29 and Ser248 are located at the interface between the N lobe (blue) and the C lobe (green) in the inward-facing structure of MurJ_{TA} (c), suggesting that adducts at these positions would trap the protein in the outward-facing state (d) by sterically blocking transition to the inward-facing state. (e) Proposed rocker-switch model of lipid II flipping by MurJ. The substrate is captured in the central cavity between the N lobe and the C lobe. Rocker-switch motion of both lobes translocates lipid II across the membrane, at which point the lipid II is released. The hydrophobic groove formed by TMs 13 and 14 (brown) of MurJ facilitates capture of the undecaprenyl tail, which remains associated with the groove during the transition to the outward-facing state. P, M, and G denote phosphate, MurNAc, and GlcNAc, respectively. Cyan circles denote the pentapeptide.

of its strategic location at the start of the hinge helix TM 7, it might influence gating between the two alternate states. The importance of the chloride or zinc ions in lipid II flipping remains to be tested.

Our studies show that the proximal site and membrane portal are critical for MurJ_{TA} function. Importantly, this region of the protein is formed by TMs 1, 2, and 8, where the structural difference is most substantial compared with the MATE structures (Supplementary Fig. 5a). It is possible that the observed structural difference in these three TMs exists to accommodate the large lipid II head group. The first two TMs in the MVF and OLF flippase families are amphipathic¹³, in agreement with the idea that the role of these TMs is to recognize the diphosphate and sugar moieties of the substrates. Furthermore, because of the arginine residues located in these three TMs, the cavity of MurJ_{TA} is mainly positively charged—unlike in typical MATE transporters—consistent with the idea that MurJ_{TA} has evolved to bind the large and negatively charged lipid II molecule (Supplementary Fig. 5b). Alternatively, this structural difference could be due to the differences between MurJ and MATEs in their conformational states or potential ion-coupling mechanisms.

On the basis of the structure of MurJ_{TA} and the MTSES functional data on *E. coli* MurJ¹⁰, we propose a rocker-switch model of lipid II flipping by MurJ (Fig. 4e). The substrate is captured in the central cavity between the N lobe and the C lobe. The rocker-switch motion is generated about a hinge on TM 7 to translocate lipid II across the membrane, after which the lipid II is released. The hydrophobic groove formed by TMs 13 and 14 of MurJ facilitates capture of the undecaprenyl tail, which remains associated with the groove during transition to the outward-facing state. Our model of lipid II flipping is strikingly different from that proposed for the lipid-linked oligosaccharide flippase PglK, for which only the outward-facing state is necessary for flipping¹⁶, indicating diverse mechanisms of lipid flipping. As it is thought that all MOP transporters use a shared transport mechanism¹³, this rocker-switch motion might be used by all MOP transporters for alternating-access transport.

METHODS

Methods, including statements of data availability and any associated accession codes and references, are available in the online version of the paper.

Note: Any Supplementary Information and Source Data files are available in the online version of the paper.

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AUTHOR CONTRIBUTIONS

A.C.Y.K. and S.-Y.L. conceived the project. A.C.Y.K. expressed, purified, crystallized, and solved the structure of MurJ_{TA} under the guidance of S.-Y.L. E.H.M. carried out the functional complementation assays under the guidance of S.-Y.L. A.C.Y.K. and S.-Y.L. wrote the manuscript.

COMPETING FINANCIAL INTERESTS

The authors declare no competing financial interests.

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- Bugg, T.D., Braddick, D., Dowson, C.G. & Roper, D.I. Bacterial cell wall assembly: still an attractive antibacterial target. *Trends Biotechnol.* **29**, 167–173 (2011).
- de Kruijff, B., van Dam, V. & Breukink, E. Lipid II: a central component in bacterial cell wall synthesis and a target for antibiotics. *Prostaglandins Leukot. Essent. Fatty Acids* **79**, 117–121 (2008).
- Ruiz, N. Lipid flippases for bacterial peptidoglycan biosynthesis. *Lipid Insights* **8**, 21–31 (2016).
- Ruiz, N. Filling holes in peptidoglycan biogenesis of *Escherichia coli*. *Curr. Opin. Microbiol.* **34**, 1–6 (2016).
- Mohammadi, T. *et al.* Identification of FtsW as a transporter of lipid-linked cell wall precursors across the membrane. *EMBO J.* **30**, 1425–1432 (2011).
- Mohammadi, T. *et al.* Specificity of the transport of lipid II by FtsW in *Escherichia coli*. *J. Biol. Chem.* **289**, 14707–14718 (2014).
- Ruiz, N. Bioinformatics identification of MurJ (MviN) as the peptidoglycan lipid II flippase in *Escherichia coli*. *Proc. Natl. Acad. Sci. USA* **105**, 15553–15557 (2008).
- Meeske, A.J. *et al.* MurJ and a novel lipid II flippase are required for cell wall biogenesis in *Bacillus subtilis*. *Proc. Natl. Acad. Sci. USA* **112**, 6437–6442 (2015).
- Inoue, A. *et al.* Involvement of an essential gene, *mviN*, in murein synthesis in *Escherichia coli*. *J. Bacteriol.* **190**, 7298–7301 (2008).
- Sham, L.-T. Bacterial cell wall. MurJ is the flippase of lipid-linked precursors for peptidoglycan biogenesis. *Science* **345**, 220–222 (2014).
- Mohamed, Y.F. & Valvano, M.A. A *Burkholderia cenocepacia* MurJ (MviN) homolog is essential for cell wall peptidoglycan synthesis and bacterial viability. *Glycobiology* **24**, 564–576 (2014).
- Meeske, A.J. *et al.* SEDS proteins are a widespread family of bacterial cell wall polymerases. *Nature* **537**, 634–638 (2016).
- Hvorup, R.N. *et al.* The multidrug/oligosaccharidyl-lipid/polysaccharide (MOP) exporter superfamily. *Eur. J. Biochem.* **270**, 799–813 (2003).

14. Butler, E.K., Davis, R.M., Bari, V., Nicholson, P.A. & Ruiz, N. Structure-function analysis of MurJ reveals a solvent-exposed cavity containing residues essential for peptidoglycan biogenesis in *Escherichia coli*. *J. Bacteriol.* **195**, 4639–4649 (2013).
15. Butler, E.K., Tan, W.B., Joseph, H. & Ruiz, N. Charge requirements of lipid II flippase activity in *Escherichia coli*. *J. Bacteriol.* **196**, 4111–4119 (2014).
16. Perez, C. *et al.* Structure and mechanism of an active lipid-linked oligosaccharide flippase. *Nature* **524**, 433–438 (2015).
17. He, X. *et al.* Structure of a cation-bound multidrug and toxic compound extrusion transporter. *Nature* **467**, 991–994 (2010).
18. Lu, M., Radchenko, M., Symersky, J., Nie, R. & Guo, Y. Structural insights into H⁺-coupled multidrug extrusion by a MATE transporter. *Nat. Struct. Mol. Biol.* **20**, 1310–1317 (2013).
19. Lu, M. *et al.* Structures of a Na⁺-coupled, substrate-bound MATE multidrug transporter. *Proc. Natl. Acad. Sci. USA* **110**, 2099–2104 (2013).
20. Tanaka, Y. *et al.* Structural basis for the drug extrusion mechanism by a MATE multidrug transporter. *Nature* **496**, 247–251 (2013).
21. Mousa, J.J. *et al.* MATE transport of the *E. coli*-derived genotoxin colibactin. *Nat. Microbiol.* **1**, 15009 (2016).

ONLINE METHODS

Protein expression and purification. The gene expressing full-length MurJ from *T. africanus* was synthesized (Bio Basic Inc.) and cloned into a modified pET26 expression vector encoding maltose-binding protein (MBP) and polyhistidine tags flanked by PreScission protease cleavage sites (PPXs), such that the final construct was His₁₀-MBP-PPX-MurJ-PPX-His₁₀. A plate of transformed C41(DE3) cells was used to inoculate 4 L of Terrific Broth. IPTG was added to a concentration of 0.4 mM once an optical density (OD₆₀₀) of 0.8 had been reached. Protein was overexpressed for 3 h at 37 °C with shaking. Cell pellets from each liter of culture were resuspended in 10 mL of lysis buffer (50 mM Tris-HCl, pH 8.0, 150 mM NaCl) containing the following additives: 10 mM β-mercaptoethanol, 1 μg/mL leupeptin, 1 μg/mL pepstatin, 5 mU/mL aprotinin, 20 μg/mL DNase I, and 2.5 mM phenylmethylsulfonyl fluoride. Cells were disrupted with a Microfluidizer, and 2.5 mM phenylmethylsulfonyl fluoride was added. Membrane proteins were extracted by stirring with 30 mM *n*-dodecyl-β-D-maltopyranoside (DDM; Anatrace) for 1 h at 4 °C. After centrifugation at 20,000g for 40 min, at 4 °C the supernatant was rotated with 1.5 mL of TALON cobalt resin (Clontech) for 1 h at 4 °C in the presence of 5 mM imidazole. Resin was recovered by centrifugation at 800g for 5 min and resuspended in 7.5 mL of buffer A (50 mM Tris-HCl, pH 8.0, 500 mM NaCl, 15 mM imidazole, 1 mM DDM). The slurry was poured into a gravity column and subsequently washed twice with 7.5 mL of buffer B (50 mM Tris-HCl, pH 8.0, 150 mM NaCl, 15 mM imidazole, 1 mM DDM). Protein was eluted in 6 mL of buffer B with imidazole added to 200 mM, and incubated overnight at 4 °C with 40 μg/mL PreScission protease and 5 mM β-mercaptoethanol. PreScission-protease-treated MurJ was purified by gel filtration on a Superdex 200 10/300 column in 20 mM Tris-HCl, pH 8.0, 150 mM NaCl, 2 mM dithiothreitol, and 0.3 mM decyl maltose neopentyl glycol (Anatrace). For preparation of selenomethionine (SeMet)-labeled protein, the same plasmid was transformed into B834(DE3) cells, and protein was overexpressed by autoinduction in the presence of 125 mg/L L-selenomethionine (ACROS Organics) according to an established protocol²² and was subsequently purified as described above.

Crystallization. We crystallized MurJ_{TA} in lipidic cubic phase (LCP) according to an established protocol²³, with some modifications. Briefly, purified MurJ_{TA} was concentrated to 15 mg/mL with a 50-kDa molecular-weight-cutoff centrifugal filter (EMD Millipore). Tris(2-carboxyethyl)phosphine (Thermo Fisher) was added to 2 mM, and protein was centrifuged at 16,000g for 30 min at 4 °C before being mixed with molten mono-olein (Sigma) in a 2:3 (w/w) water:lipid ratio with a twin-syringe setup. MurJ-reconstituted LCPs were dispensed on 96-well glass sandwich plates in 150-nL drops by a Gryphon LCP robot (Art Robbins Instruments) and overlaid with 1 μL of precipitant solution. Both native and SeMet MurJ_{TA} crystallized at room temperature in solutions containing 30–45% (v/v) PEG 400, 200–400 mM CaCl₂, pH 6.5–9.0. Crystals grew to full size in 4 weeks and were flash-frozen in liquid nitrogen without additional cryo-protectant.

Addition of lipid II improved diffraction quality and was performed for native crystals as follows: the L-lysine form of lipid II (BaCWAN, University of Warwick) was added to mono-olein powder in a glass vial to a final concentration of 1.5 mM (in molten mono-olein). The mono-olein powder was dissolved in 1 mL of chloroform and mixed thoroughly. This mixture was dried under an argon stream and washed with 0.5 mL of pentane. Finally, organic solvent was removed by complete drying under an argon stream before the lipids were melted in a 42 °C water bath. LCP trays were set up as described above.

For the bromide soak experiment, a window was cut from the coverslip of selected wells, and the precipitant solution was wicked away. Fresh precipitant solution (of the same composition except with CaCl₂ replaced by CaBr₂) was injected and the well was resealed with tape. Soaking lasted for 2 h, after which crystals were immediately harvested and frozen.

Data collection and structure determination. The structure of MurJ was determined by single-wavelength anomalous dispersion (SAD). Data were collected at the NECAT 24-ID-C and 24-ID-E beamlines (Advanced Photon Source, Argonne National Laboratory) with a wavelength of 0.9791 Å and were processed by HKL2000 (HKL Research). Because of the high isomorphism between crystals, selected frames from 12 SeMet crystals were merged together in HKL2000 to generate a data set with over 50-fold average redundancy and signal extending to 3.2-Å resolution. Heavy-atom sites were located by SHELXD²⁴, and

phasing was performed by PHENIX AutoSol²⁵ initially to a high-resolution cutoff of 5.0 Å, producing a figure of merit of 0.50. We gradually extended phases to 3.5 Å using MR-SAD by placing idealized helical fragments into the density and density modification (Supplementary Fig. 1). The eight observed Se sites, corresponding to all the methionine residues in the protein except the initiator, were sufficient for unambiguous assignment of topology and register (Supplementary Fig. 1). For the high-resolution data, we merged selected frames from three native crystals crystallized in the presence of lipid II to generate a data set with signal extending to 2.0-Å resolution. Phases were obtained from the model built from the SeMet data by PHASER²⁶ and were improved by cycles of model-building in COOT and refinement in PHENIX.refine to a final $R_{\text{work}}/R_{\text{free}}$ of 18.9/21.4 (Table 1), good geometry (98% Ramachandran favored, 0% Ramachandran outliers), and minimal clashes (clashscore of 1). MurJ_{TA} crystallized in space group C2 with a single MurJ molecule in the asymmetric unit. The quality of the electron density map (Supplementary Fig. 1d) was sufficient to enable us to unambiguously build residues 4–470 (out of a total of 475 residues), as well as to place a shell of ordered mono-olein and water molecules around the protein. For the bromide soak experiment, data were collected at beamline 24-ID-C with a wavelength of 0.9197 Å. Phases were obtained from the high-resolution model by PHASER, and the anomalous difference Fourier map was calculated by PHENIX.maps.

Docking of lipid II and structure analysis. Docking of lipid II to the structure of MurJ_{TA} was done with Autodock Vina²⁷. We chose the L-lysine form of lipid II for docking instead of the meso-diaminopimelate (mDAP) form because the related bacterium *Thermotoga maritima* was shown to incorporate both L- and D-lysine into peptidoglycan instead of mDAP^{28,29}. Initial lipid II coordinates were generated from 2D geometry in PHENIX.eLBOW, and stereochemistry was manually corrected in PHENIX.REEL, with reference to the NMR structure of lipid II in complex with nisin (PDB 1WCO)³⁰. In particular, effort was made to ensure that all 16 chiral centers had the correct R/S configuration, that the peptide and polyprenyl C=C bonds had the right *cis/trans* configuration and planarity, and that the sugars had the correct pucker and glycosidic linkage. The remaining 68 torsions were freely rotatable during docking. In addition, the side chains of Arg18 and Glu57 of MurJ_{TA} were treated as flexible for docking, as they appeared to be flexible in our structure (high B factors as compared with side chains of adjacent residues). Docking was carried out over a search space of 30 × 30 × 36 Å covering the central cavity, portal, and hydrophobic groove. Surface electrostatics were calculated with the APBS³¹ PyMOL plugin. Molecular graphics were created in PyMOL³² and CueMol (<http://www.cuemol.org/>). Sequence alignment was done with structure restraints by PROMALS3D³³, and structure superposition was done in UCSF Chimera³⁴.

MurJ complementation assay. MurJ_{TA} was cloned into pEXT21 with a C-terminal Flag tag and was subjected to site-directed mutagenesis using standard recombinant DNA methods. Expression of MurJ_{TA} is controlled by the tac promoter in pEXT21 (ref. 11). The pEXT21 plasmid bearing wild-type MurJ_{TA} or a mutant thereof was transformed into the conditional MurJ-depletion *E. coli* strain NR1154 (ref. 7), which was a gift from Dr. Natividad Ruiz (The Ohio State University). The endogenous MurJ in NR1154 is placed under the control of the P_{BAD} promoter; therefore, NR1154 growth depends on the presence of L-arabinose. D-fucose is an anti-inducer of endogenous *E. coli* MurJ expression in NR1154. Transformants were selected with 80 μg/mL spectinomycin on LB-agar plates supplemented with 0.2% arabinose. Overnight cultures were diluted to an OD₆₀₀ of 0.2 and grown to an OD₆₀₀ of 0.6 in the presence of 80 μg/mL spectinomycin and 0.2% arabinose. IPTG was then added to the duplicate of each culture at a final concentration of 0.1 mM. Cultures were further incubated with shaking for 3 h at 37 °C before cells were harvested by centrifugation. For the complementation assay, cells were washed three times with PBS and normalized to an OD₆₀₀ of 0.5. Each culture was then diluted to 0.02, subjected to ten-fold serial dilution, and spotted (5 μL) onto LB-agar plates supplemented with either 0.05% fucose or 0.05% fucose with 0.1 mM IPTG.

For western blotting to check for protein expression, cells were lysed with a Microfluidizer, and total membrane was isolated by ultracentrifugation. We prepared samples by resuspending the membrane in 50 mM Tris-HCl, pH 8, 150 mM NaCl, mixing it with a loading buffer (final concentrations of 3 M urea, 2.5% SDS, 60 mM Tris-HCl, pH 6.8, 0.01% bromophenol blue, 1 mM β-mercaptoethanol),

and subjecting it to 3 min of sonication before loading it onto a 10% SDS-PAGE gel. Bands were transferred to a PVDF (polyvinylidene difluoride) membrane, which was probed with mouse anti-Flag primary antibody (1:1,200 dilution, Sigma, F3165; antibody-validation information is available at <http://www.sigmaaldrich.com/catalog/CertOfAnalysisPage.do?symbol=F3165&LotNo=slbn8915v>) followed by donkey anti-mouse secondary antibody (1:10,000 dilution, LI-COR, 926-32212). Results were visualized on a LI-COR Odyssey CLx imaging system.

Data availability. Atomic coordinates and structure factors for the reported crystal structure have been deposited in the Protein Data Bank under accession code [5T77](#). Atomic coordinates for the outward-facing model are included as **Supplementary Data Set 1**. Any other data pertaining to this paper are available upon request.

22. Fox, B.G. & Blommel, P.G. Autoinduction of protein expression. *Curr. Protoc. Protein Sci.* Chapter 5, Unit 5.23 (2009).
23. Caffrey, M. & Cherezov, V. Crystallizing membrane proteins using lipidic mesophases. *Nat. Protoc.* **4**, 706–731 (2009).
24. Sheldrick, G.M. A short history of SHELX. *Acta Crystallogr. A* **64**, 112–122 (2008).
25. Adams, P.D. *et al.* PHENIX: a comprehensive Python-based system for macromolecular structure solution. *Acta Crystallogr. D Biol. Crystallogr.* **66**, 213–221 (2010).
26. McCoy, A.J. *et al.* Phaser crystallographic software. *J. Appl. Crystallogr.* **40**, 658–674 (2007).
27. Trott, O. & Olson, A.J. AutoDock Vina: improving the speed and accuracy of docking with a new scoring function, efficient optimization, and multithreading. *J. Comput. Chem.* **31**, 455–461 (2010).
28. Huber, R. *et al.* *Thermotoga maritima* sp. nov. represents a new genus of unique extremely thermophilic eubacteria growing up to 90 °C. *Arch. Microbiol.* **144**, 324–333 (1986).
29. Boniface, A., Bouhss, A., Mengin-Lecreux, D. & Blanot, D. The MurE synthetase from *Thermotoga maritima* is endowed with an unusual D-lysine adding activity. *J. Biol. Chem.* **281**, 15680–15686 (2006).
30. Hsu, S.T. *et al.* The nisin–lipid II complex reveals a pyrophosphate cage that provides a blueprint for novel antibiotics. *Nat. Struct. Mol. Biol.* **11**, 963–967 (2004).
31. Baker, N.A., Sept, D., Joseph, S., Holst, M.J. & McCammon, J.A. Electrostatics of nanosystems: application to microtubules and the ribosome. *Proc. Natl. Acad. Sci. USA* **98**, 10037–10041 (2001).
32. Delano, W.L. *The PyMOL Molecular Graphics System* (DeLano Scientific, 2002).
33. Pei, J., Kim, B.H. & Grishin, N.V. PROMALS3D: a tool for multiple protein sequence and structure alignments. *Nucleic Acids Res.* **36**, 2295–2300 (2008).
34. Pettersen, E.F. *et al.* UCSF Chimera—a visualization system for exploratory research and analysis. *J. Comput. Chem.* **25**, 1605–1612 (2004).